

ASSIGNMENT #2

Course: CHE 301. CHEMICAL ENGINEERING THERMODYNAMICS
Term: Spring 2020
Professor: Dr. Michael Caracotsios
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Your name: _____

Due Date: Wednesday February 05 by 5:00pm in my office

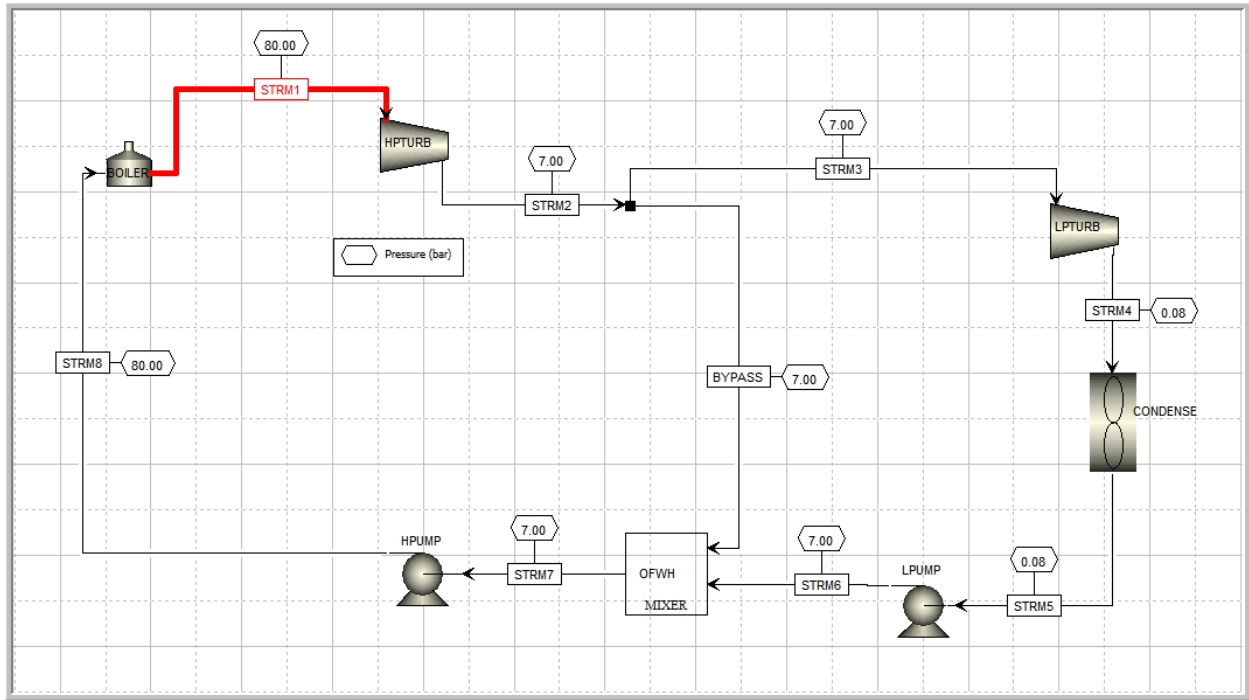
Number of points.....100

Your score.....

Note: All problems are worth 100 points. Your final score will be the average value

Problem 1A: Regenerative Rankine Cycle

1000 MMSCFD of superheated steam generated in a boiler at $P_0 = 80 \text{ bar}$ and $T_0 = 480^\circ \text{C}$ are expanded into a steam condensing turbine (CST) to a pressure $P_2 = 8 \text{ kPa}$. To improve the efficiency of the power cycle a bypass stream is mixed with the discharge of the low pressure pump of the condenser effluent in order to preheat the boiler feed water. (See schematic below known as **Regenerative Rankine Cycle**).



If the turbine has an efficiency of 88% and the feed water pumps efficiency of 80% calculate:

1. The maximum flow rate of the steam bypass in **MMSCFD**
2. The amount of power generated by the cycle in **MW**
3. The efficiency of the cycle
4. The amount of water in **gpm** required for the condenser, assuming a water temperature rise of 11°C
5. If the power plant is using coal as fuel in the boiler give an estimate of the amount of coal needs in **tonne/day**
6. If the power plant is equipped with cooling tower, how much water does it draw from the environment, and how much it discharges as waste. Express your answers in **gpm**

You may use the **CHE301 Calculator.xlsm** for your calculations. If your Excel does not support the steam calculations use **AthenaVisual Studio** or any other available resource to obtain the steam/water thermodynamic properties.

Place all your answers in boxes for easy readability. Clearly state your assumptions and references for additional data you may need to answer the questions.